

Mentoring Plan

The project supports one postdoctoral researcher for four years, co-advised by PI Napolitano (structural vulnerability and community co-design), Co-PI Wauthier (SAR processing and uncertainty quantification), and Co-PI Busse (hydrological modeling and mass balance). All PIs are at Penn State, enabling continuous cross-group engagement rather than periodic visits. The postdoc spans the project's technical lanes, implementing SAR data processing and Bayesian fusion under Wauthier's direction, building the mass balance model under Busse's direction, and contributing to fragility calibration and community interface development under Napolitano's direction. The postdoc receives cross-disciplinary training spanning SAR imagery processing, hydrological mass balance modeling, probabilistic structural assessment, and community-engaged research. The plan will be updated annually based on mentee feedback and assessed through progress reports and professional achievements.

Regular meetings and cross-disciplinary co-mentoring: PI Napolitano serves as primary advisor and holds weekly meetings with the postdoc. Co-PIs Wauthier and Busse each hold biweekly one-on-one meetings aligned with their technical lanes. Monthly cross-team meetings ensure integration across the SAR, hydrological, and co-design components. The postdoc attends lab meetings across all three groups: Wauthier's SAR sessions, Busse's mass balance meetings, and Napolitano's structural assessment and community engagement meetings. The postdoc participates in Napolitano's annual community co-development workshops in Montpellier during Years 1–3.

Career counseling and professional development: The postdoc will present at conferences (AGU, ASCE EMI, Association for Preservation Technology) and submit to peer-reviewed journals. Each PI provides career counseling on academic, industry, and government pathways.

Grant writing and scholarly communication: Training in preparing NSF proposals, publications, and presentations through Penn State's Graduate Writing Center. PI Napolitano will share materials from her graduate research methods course. The postdoc will develop at least one independent funding application during the appointment.

Teaching and mentoring skill development: The postdoc will co-supervise BESURE undergraduate researchers each year, developing supervision and mentoring skills. The postdoc will receive mentorship on undergraduate supervision through the Penn State Undergraduate Research and Fellowships Mentoring program and is encouraged to guest lecture in PI courses and attend pedagogy workshops through the CIRTL Network.

Undergraduate research: Two undergraduates per year participate through the BESURE program at Penn State, with recruitment prioritizing underrepresented groups via the WISER and MURE programs. The postdoc serves as day-to-day mentor for these students.

Diversity training and responsible practices: The postdoc participates in DEI training at Penn State and completes responsible conduct of research training through Penn State's SARI program, covering research ethics, data management, and intellectual property.